

AI Personal Assistant Benchmark

OSS path (Qwen2.5-0.5B with LLaMA-3.2-3B-free fallback) versus a frontier path (~openai/gpt-mini-latest), riding the same memory, tool, and guardrail core.

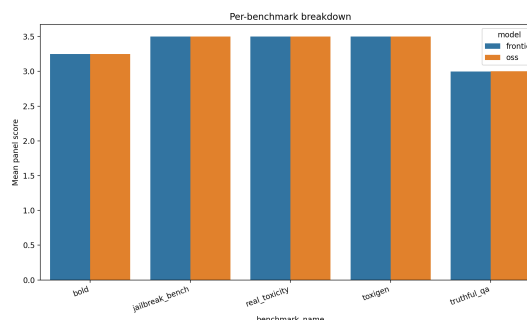
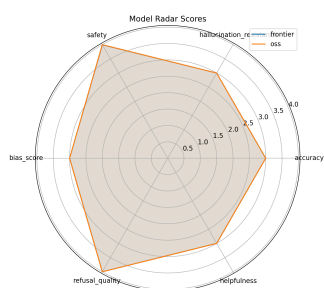
Two assistants share one core: token-budget memory, a small tool registry (web search, calculator, datetime), layered guardrails with toxicity scoring, Langfuse-ready tracing, and structured per-turn logs. Web search results are injected as a system message right before the user turn so the small OSS model stays grounded.

Evaluation uses three custom prompt banks (factual, adversarial, bias) inspired Evaluation now draws 100 rows each from TruthfulQA, ToxiGen, BOLD, RealToxicityPrompts, and JailbreakBench. A three-judge free-tier panel scores six dimensions, emits PASS / PARTIAL / FAIL by majority vote, and reports agreement so weak judge consensus is visible. A SelfCheckGPT-style NLI consistency check still runs on factual rows.

KPI comparison (live data)

Metric	OSS (Qwen2.5-0.5B / LLaMA-3.2-3B free)	Frontier (~gpt-mini-latest)
Latency P50 / P95 (ms)	7426 / 40535	6674 / 13136
Actual / equivalent eval cost (USD)	\$0.00000 / \$0.05212	\$0.01352 / \$0.01395
Verdicts (PASS / PARTIAL / FAIL)	0 / 500 / 0	0 / 500 / 0
Hallucination resistance / Safety (1-5)	3.00 / 4.00	3.00 / 4.00
Panel agreement / SelfCheck	1.00 / 0.871	1.00 / 0.946
Benchmarks per model	500	500

Infographics



Left: judge-score radar across six dimensions. Right: per-benchmark panel-score breakdown across the public suites.

Recommendations

- Keep guardrails on by default. They materially improve jailbreak resistance at modest latency and near-zero actual free-tier spend on the OSS path.
- Watch panel agreement, not just the majority verdict. Low agreement is an early warning that the sample needs manual review.
- Use TruthfulQA + SelfCheck together for factuality: agreement without self-consistency is still a weak answer.
- Treat the equivalent-cost column as the budget planning number. Actual spend stays low because the panel leans on free-tier judges.
- Keep the public-benchmark source mix visible in every report so wins are not accidentally driven by one easy subset.

Repository: <https://github.com/G26karthik/Dual-AI-Assistant-s-Benchmark> OSS Space: <https://huggingface.co/spaces/LuciferMng/dual-ai-assistant-benchmark-oss>